

STAT 420/626 Course Project Proposal

Robust Brain Tumor Segmentation with Missing MRI Modalities

Jing Wang

Yechen Liu

Thomas Duong

September 25, 2026

Abstract

We propose a controlled study of brain tumor segmentation when an MRI sequence is unavailable. Using the labeled BraTS 2023 cohort, we will compare a lightweight 2.5D U-Net, the same network trained with modality dropout, and a version that also encourages agreement between complete- and incomplete-input predictions. Patient-level data splitting, matched training budgets, and paired confidence intervals will support an assessment of robustness and computational cost. The contribution is a reproducible empirical study of an existing modeling idea, with particular attention to the additional value of consistency regularization. All experiments described below are planned; no results are claimed.

1 Background

1.1 What data science problem will we solve?

Our task is voxel-level segmentation of glioma subregions in multimodal brain MRI. We will predict whole tumor (WT), tumor core (TC), and enhancing tumor (ET). The central question is: consistency regularization improve a modality-dropout U-Net when one MRI modality is missing, while preserving useful performance with complete inputs?. This is a segmentation study in patients with glioma, rather than a patient-level screening study. The selected cohort does not provide a healthy-control comparison for evaluating tumor-versus-no-tumor diagnosis [1].

1.2 Why is this problem important?

Delineating tumor extent provides objective, reproducible measurements that can support treatment planning and longitudinal disease monitoring [4]. The BraTS benchmark offers expert-approved tumor-subregion annotations for studying this task [1–3]. A model developed with complete multimodal inputs may be less useful when an acquisition is absent; prior work directly addresses missing sequences through channel/sequence dropout and self-distillation [5–7]. For our statistics course, the problem connects conditional probability estimation, regularization, dependent observations, and uncertainty in model comparisons. Artificially removing sequences gives a controlled robustness experiment, although it cannot establish clinical effectiveness.

1.3 How will we formulate the model?

Let X contain four co-registered MRI volumes and let Y be the voxel-label map. Let m be a four-entry binary indicator of available modalities. We will fit a conditional discriminative model

$$p(Y \mid X_{\text{obs}}, m),$$

where X_{obs} denotes the observed subset of modalities and the softmax outputs estimate four class probabilities per voxel. CNNs and vision transformers are both valid architecture choices for this discriminative task. A generative alternative would instead model the conditional distribution of the missing images,

$$p(X_{\text{miss}} \mid X_{\text{obs}}),$$

which could be used for imputation or data augmentation. We defer that generative direction to future work because it requires additional modeling assumptions and compute, and because our primary question concerns segmentation robustness under missingness [8]. The consistency term in our proposed model (M2) regularizes $p(Y \mid X_{\text{obs}}, m)$ by encouraging agreement between the complete-view and missing-view predictions, thereby promoting robustness to the modality-availability pattern m .

2 Data Description

2.1 Dataset availability and study population

We will use the **ASNR-MICCAI BraTS 2023 Adult Glioma (GLI) training cohort** [1]. Each case contains four co-registered MRI modalities: T1, contrast-enhanced T1 (T1Gd/T1ce), T2, and T2-FLAIR, along with a segmentation mask. Images are distributed as NIfTI volumes with official preprocessing already performed, including re-orientation, co-registration, resampling to 1 mm isotropic spacing, and skull-stripping. We will follow the published benchmark definitions for the Adult Glioma segmentation task.

We have obtained authorized access to the dataset and have downloaded the BraTS 2023 GLI training cohort (file name: **ASNR-MICCAI-BraTS2023-GLI-Challenge-TrainingData.zip**). The official BraTS 2023 data is available through Synapse following registration and acceptance of the data usage agreement [1]. The training partition contains **1,251 labeled cases** with expert annotations for three tumor subregions: enhancing tumor (ET), tumor core (TC), and whole tumor (WT) [1]. The planned upper bound is 1,251 cases; after quality control we will report the exact retained sample size N and document any exclusions. To preserve a fully labeled held-out evaluation set under our own control, we will create patient-level training, validation, and test splits from this labeled training cohort.

A mapping file (**BraTS2023_2017_GLI_Mapping.xlsx**) is also included in our downloaded folder, which allows cross-referencing between BraTS 2023 and earlier BraTS releases. We will use this only for documentation and will not mix releases to avoid data leakage between challenge years.

2.2 Cleaning, missing data, and retained sample size

We will check file readability, required modalities, finite intensities, consistent orientation, spacing, dimensions, label values, and duplicate patient identifiers. We will inspect representative image-mask

overlays and log every exclusion with a reason. Unusual tumor size or appearance alone will not be an exclusion criterion. A case with an irreparably corrupt or absent required file will be excluded from the complete-input experimental cohort; naturally incomplete cases, if encountered, will be counted separately.

The retained size is $N_{\text{clean}} = 1251$ minus documented exclusions; 1,251 is the official training partition size, not an observed post-cleaning count. For each available modality, we will z-score nonzero brain voxels using that volume’s own mean and standard deviation, with a small numerical safeguard. This operation uses no labels and can also be applied to an unseen scan. Original background zeros will remain zero. Any population-level preprocessing parameters will be estimated from training patients only. Artificially missing channels will be set to zero after normalization and identified by availability indicators; zero filling is a missingness code, not anatomical imputation.

2.3 Input variables and leakage prevention

For each central axial slice, we will stack that slice and its two immediate neighbors from each of the four modalities: 12 image channels. Four constant availability maps bring the input to 16 channels. Boundary slices will repeat the nearest available slice. Missing one modality removes all three of its image channels. Spatial augmentation will be identical across image channels and the central label mask. Age, survival outcomes, filenames, and patient identifiers will not be predictors.

Before extracting slices, we will split patients approximately 70%/15%/15% into training, validation, and held-out testing, using a fixed split seed of 626. If all 1,251 cases are retained, the target counts are approximately 876/188/187. We will stratify by whether ET is present when feasible. All scans and derivatives of one patient remain in one partition. We will use only this BraTS 2023 GLI release to avoid overlap between challenge years.

2.4 Expected outputs and labels

The stored segmentation masks are mutually exclusive, four-class label maps: 0 denotes background, 1 denotes necrotic tumor core (NCR), 2 denotes peritumoral edematous/invaded tissue (ED), and 3 denotes enhancing tumor (ET). BraTS 2023 GLI uses label 3, rather than the legacy label 4, for ET [1]. For four-class cross-entropy, the stored labels are used directly as class indices $\{0, 1, 2, 3\}$; predictions are mapped back to the original label values for evaluation. We will derive the standard composite regions from the hard class map: $\text{WT} = \{1, 2, 3\}$, $\text{TC} = \{1, 3\}$, and $\text{ET} = \{3\}$. Each central slice is extracted together with its corresponding label slice from the patient-level mask. Cropping, padding, and geometric augmentation are applied identically to images and masks, using nearest-neighbor interpolation for masks. Removing a modality changes only the input view, never the target label map.

For each held-out patient and each input condition, a model will produce four voxelwise softmax probabilities and an argmax hard-label map. We will reconstruct central-slice predictions into a patient-level segmentation and retain the probability maps, hard masks for each input condition (complete and the four single-missing cases), WT/TC/ET masks, Dice and HD95 values, inference time, peak GPU memory, and model parameter count. We will also save the patient split, random seed, missing-modality indicator, selected λ , and training checkpoint so that every reported comparison can be reproduced. These outputs quantify segmentation robustness and will support the paired comparisons described in Section 4; they are not a tumor-versus-no-tumor diagnostic score.

3 Modeling

3.1 Is there any related work that provides methodologies for our problem?

U-Net is a foundational model for medical image segmentation. It uses an encoder, a decoder, and skip connections to locate structures at the voxel level [9]. nnU-Net builds on this by automatically choosing preprocessing, architecture, and training settings, and it performs well on many tasks [10]. TransBTS adds transformer layers to a 3D CNN to capture longer-range context in brain tumor images [11].

Several methods directly address missing MRI sequences. Shen and Gao use channel dropout and feature-level domain adaptation to support absent modalities [5]. Feng et al. evaluate sequence dropout and show that it can improve robustness to missing sequences without a statistically detectable loss for complete inputs [6]. M3AE uses modality masking and self-distillation across missing-modality states [7]. Generative modality-completion methods are another option; for example, U-HVED jointly completes missing modalities and segments tumors [8]. Therefore, modality dropout and consistency are established ideas. The contribution of this project is to isolate the added value of a simple complete-to-missing consistency loss in a matched, low-compute 2.5D U-Net comparison.

3.2 What are the limitations of existing methods?

Models trained only on complete inputs, such as U-Net or nnU-Net [9, 10], are not built to handle missing sequences, and their benchmark scores do not show how robust they are. 3D transformer models [11] and generative imputation [8] need a lot of memory and compute, which is hard for a course project. Missing-modality methods [5–7] usually combine several parts at once and are tested with different data splits and training budgets, so it is hard to tell how much a consistency term helps on its own. For example, M3AE already contains a related self-distillation idea [7], but it is combined with masked-autoencoder pretraining and other components; our contribution is a controlled comparison that isolates its incremental effect. Also, forcing the missing-input prediction to match the complete-input prediction may hurt when the missing sequence has information the others cannot replace, such as T1ce for enhancing tumor.

3.3 What is our plan to improve existing methods?

Instead of proposing a brand-new method, we run a fair, low-compute comparison to see whether consistency regularization adds value beyond modality dropout. All models share one 2.5D U-Net with four encoder levels (32, 64, 128, 256 channels), a 512-channel bottleneck, skip connections, instance normalization, ReLU, and a four-class softmax output. The model predicts the central slice using its neighbors, which gives some 3D context without the memory cost of a full 3D model.

We train three models with the same architecture and training budget. M0 uses only complete inputs. M1 follows channel/sequence dropout approaches [5, 6] and also trains on a second view with one random modality removed. M2, our proposed model, adds a consistency loss related to missing-modality self-distillation [7] that makes the missing-input prediction agree with the complete-input prediction. The segmentation loss combines cross-entropy and soft Dice over the

three tumor classes:

$$\mathcal{L}_{\text{seg}}(p, y) = \mathcal{L}_{\text{CE}}(p, y) + 1 - \frac{1}{3} \sum_{c=1}^3 \frac{2 \sum_v p_{vc} y_{vc} + \epsilon}{\sum_v p_{vc} + \sum_v y_{vc} + \epsilon}. \quad (1)$$

M2 minimizes

$$\mathcal{L} = \frac{1}{2} [\mathcal{L}_{\text{seg}}(p^{\text{full}}, y) + \mathcal{L}_{\text{seg}}(p^{\text{miss}}, y)] + \lambda \mathcal{L}_{\text{cons}}, \quad \mathcal{L}_{\text{cons}} = \frac{1}{|\Omega|} \sum_{v \in \Omega} D_{\text{KL}}(\text{sg}(p_v^{\text{full}}) \| p_v^{\text{miss}}), \quad (2)$$

where Ω is the set of brain voxels. The complete-input prediction acts as a soft target, and setting $\lambda = 0$ gives back M1. Since this term could also hurt, we will test its effect instead of assuming it helps.

3.4 What is our plan to train the model and show that the proposed method works?

We will use PyTorch with AdamW (learning rate 10^{-4} , weight decay 10^{-5}), mixed precision, and batch size 4. Each epoch samples 32 slices per training patient, about half with tumor. We train for up to 80 epochs, check validation every five epochs, and stop early after five checks with no improvement. For M2 we try $\lambda \in \{0.01, 0.1, 1.0\}$ and choose it on the validation set using average Dice over complete inputs and the four single-missing cases. All model selection (early stopping and λ) uses validation patients only. We then train M0, M1, and M2 with three matched seeds (11, 22, 33), so that each M1 and M2 run shares the same seed, and only use the test set at the end.

We test each model with complete inputs and with T1, T1ce, T2, or FLAIR missing. We report WT, TC, and ET Dice,

$$\text{Dice}(A, B) = \frac{2|A \cap B|}{|A| + |B|}, \quad (3)$$

along with HD95, inference time, GPU memory, and parameter count. The pre-specified **primary estimand** is the patient-level mean Dice difference M2 – M1, averaged over the three regions, the four single-missing conditions, and the three seeds. We use a patient-level bootstrap with 2,000 resamples to get a 95% confidence interval. Secondary contrasts (M1 vs. M0, and each region and missing modality separately) are reported with intervals but are not used to claim success.

If the confidence interval for the primary estimand is above zero, it supports adding consistency regularization. If it includes zero, the benefit is unclear, and if it is negative we will report that. Separately, for complete inputs we compute a 95% bootstrap confidence interval for M2 – M1 and pre-define non-inferiority as its lower limit being above -0.02 ; a CI above zero is only the test for improvement under missingness. Our results use simulated missingness and one random split, so they do not show performance at new hospitals, and we will not compare them directly with published leaderboard scores.

References

- [1] ASNR-MICCAI BraTS 2023 Challenge, “Adult Glioma Challenge.” [Online]. Available: <https://www.synapse.org/Synapse:syn51156910/wiki/622351>. [Accessed: Sep. 23, 2026].
- [2] B. H. Menze et al., “The multimodal brain tumor image segmentation benchmark (BRATS),” *IEEE Trans. Med. Imaging*, vol. 34, no. 10, pp. 1993–2024, 2015, doi: 10.1109/TMI.2014.2377694.
- [3] S. Bakas et al., “Advancing The Cancer Genome Atlas glioma MRI collections with expert segmentation labels and radiomic features,” *Sci. Data*, vol. 4, Art. no. 170117, 2017, doi: 10.1038/sdata.2017.117.
- [4] F. J. Dorfner, J. B. Patel, J. Kalpathy-Cramer, E. R. Gerstner, and C. P. Bridge, “A review of deep learning for brain tumor analysis in MRI,” *NPJ Precis. Oncol.*, vol. 9, Art. no. 2, 2025, doi: 10.1038/s41698-024-00789-2.
- [5] Y. Shen and M. Gao, “Brain tumor segmentation on MRI with missing modalities,” in *Information Processing in Medical Imaging—IPMI 2019*, LNCS 11492, pp. 417–428, 2019, doi: 10.1007/978-3-030-20351-1_32.
- [6] X. Feng et al., “Brain tumor segmentation for multi-modal MRI with missing information,” *J. Digit. Imaging*, vol. 36, no. 5, pp. 2075–2087, 2023, doi: 10.1007/s10278-023-00860-7.
- [7] H. Liu, D. Wei, D. Lu, J. Sun, L. Wang, and Y. Zheng, “M3AE: Multimodal representation learning for brain tumor segmentation with missing modalities,” *Proc. AAAI Conf. Artif. Intell.*, vol. 37, no. 2, pp. 1657–1665, 2023, doi: 10.1609/aaai.v37i2.25253.
- [8] R. Dorent, S. Joutard, M. Modat, S. Ourselin, and T. Vercauteren, “Hetero-modal variational encoder-decoder for joint modality completion and segmentation,” in *Medical Image Computing and Computer Assisted Intervention—MICCAI 2019*, LNCS 11765, pp. 74–82, 2019, doi: 10.1007/978-3-030-32245-8_9.
- [9] O. Ronneberger, P. Fischer, and T. Brox, “U-Net: Convolutional networks for biomedical image segmentation,” in *Medical Image Computing and Computer-Assisted Intervention—MICCAI 2015*, LNCS 9351, pp. 234–241, 2015, doi: 10.1007/978-3-319-24574-4_28.
- [10] F. Isensee, P. F. Jaeger, S. A. A. Kohl, J. Petersen, and K. H. Maier-Hein, “nnU-Net: A self-configuring method for deep learning-based biomedical image segmentation,” *Nat. Methods*, vol. 18, pp. 203–211, 2021, doi: 10.1038/s41592-020-01008-z.
- [11] W. Wang, C. Chen, M. Ding, H. Yu, S. Zha, and J. Li, “TransBTS: Multimodal brain tumor segmentation using transformer,” in *Medical Image Computing and Computer Assisted Intervention—MICCAI 2021*, LNCS 12901, pp. 109–119, 2021, doi: 10.1007/978-3-030-87193-2_11.